

The Impact of Social Media Use on Teenager Psychological and Risk Outcomes: A Structural Equation Modeling Analysis

Abstract

This study investigates the relationships between teenagers' social media use and its psychological consequences, using data from the Pew Research Center's Teens and Technology 2018 survey (N = 743). A confirmatory factor analysis (CFA) was first conducted to validate the measurement model for each construct. Based on this, a structural equation model (SEM) was developed to examine how social media use control, online interaction, and self-awareness relate to teenagers' perceptions of positive and negative outcomes. Multiple group analysis (MGA) further explored differences based on teenagers' experiences of online harassment/bullied. The findings largely supported the proposed model, showing that active engagement and regulation of social media use were associated with more positive outcomes. Negative self-awareness was associated with negative consequences, while higher levels of online interaction unexpectedly is negatively correlated with negative self-awareness. These results highlight the complexity of social media's effects on teenagers and suggest that the quality and context of engagement are critical. Limitations of the cross-sectional design are acknowledged, and directions for future research are outlined.

Introduction

In modern society, social media has become an integral part of the lives of teenagers. They use social media to communicate with their peers, share information, and express their identity. Based on the increasing influence of social media, the academic discussion and social interests on how it affects teenagers' health, social relationships, and their well-being.

Existing studies that analyze the effects of social media on teenagers show conflicting and opposite results. For example, there are studies that emphasize positive effects such as information acquisition or feeling sense of belonging (Sikolia & Mberia, 2019; Jin-zhu, 2016). On the other hand, there are other studies that analyze the negative aspects of social media. For example, there are increased depression, decreased life satisfaction or increased negative self-awareness from the social media (Ali et al., 2025; Ginting & Sahlepi, 2024).

These conflicting results suggest that it is difficult to explain the impact based on a simple quantitative approach. Rather, qualitative aspects and personal factors, such as how to use social media, should be considered. So, the need to explore in-depth the various social media usage and

self-control abilities of teenagers and the specific pathways that lead to positive and negative outcome perception is raised.

Therefore, this study aims to investigate the structural relationship between various aspects of social media use and its results by utilizing large-scale survey data (Pew Research Center, 2018) on the social media experience of American adolescents. Specifically, this study aims to analyze the relationship between the variables through a structural equation model (SEM) and multi-group analysis (MGA).

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Literature Review

For H1, this article expects that teenagers' social media posting would increase the frequency and nature of digitally mediated social interaction and connection. This hypothesis is based on the existing theory (e.g., the hyper-personal model) that online circumstances promote self-exposure, facilitating the building of relationships. (Masaviru, 2016; Nguyen, 2012) Previous studies provide evidence to support the relationship. Self-exposure in online circumstances contributes to getting social support (Nguyen & Ho, 2022) and to increasing intimacy and interactions with friends (Desjarlais, 2020). However, this relationship is not always positive. Superficial interaction on a social media platform (Klinkhoff, 2017) or in specific individuals (e.g., teenagers with low social anxiety) or in a specific situation (e.g., excessive exposure), self-exposure rather impairs a sense of connection (Miller, 2020) or has a negative consequence such as cyber violence (Ho & Nguyen, 2022). Despite these complex aspects, considering the fundamental role of online self-expression in relationship formation and maintenance and the results of a number of studies on its positive effects, this study aims to establish and verify a hypothesis that H1: SMP will have a positive effect on PFSA.

As a second hypothesis, it expects that the higher the level of digital-mediated social interaction and connection (PFSA) of adolescents, the higher the positive outcome perception (PC) of social media use. In other words, the study assumes that the experience of actively interacting with others online will lead to positive perceptions (e.g., promoting a sense of connection, obtaining support, usefulness, etc.) of social media. The previous study supports H2 by showing that online interactions positively affect teenagers. They feel emotionally connected with other mates through interaction on the online platform and release their stress with interactive activities (O'Reilly et al., 2022). Online peer support networks are a valuable resource for coping with difficulties, and this behavior has also shown a protective effect of lowering cyber aggression (Xie & Xie, 2019). In addition, online social interaction positively affects self-esteem and identity development. Experiences of receiving positive feedback and recognition from peers can increase adolescents' self-esteem (Norton, 2018), and opportunities to experiment with identity in online spaces can be a satisfactory experience that promotes personal growth and self-exploration, especially for teenagers who are lonely or lack offline support (Leung, 2011). Based on these findings, this study verifies the hypothesis that PFSA will positively affect PC.

The third hypothesis predicts that teenagers' social media content management and concerns (SMUC) will positively impact positive outcome perception (PC). Based on the literature, SMUC lets teenagers manage their digital persona (Nyamwesa, 2024), secure their privacy, such as limiting access to their posts (Sinyavskaya, 2022), adjust audience settings (Meertens, 2014), and make a balance between the benefits and risks of social media (Nyamwesa, 2024). These various management behaviors are affected by privacy concerns, data control awareness, or privacy literacy (Wolf et al., 2023) and aim to avoid conflict and respond to perceived social threats (Meertens, 2014). Through an effective SMUC strategy, teenagers can mitigate online risks such as cyberbullying and enjoy positive benefits (Nyamwesa, 2024). Therefore, this study sets up the hypothesis that SMCU positively impacts PC.

The fourth hypothesis predicts that teenagers' levels of social media posting and self-expression (SMP) will affect their levels of content management and related concerns (SMUC). Based on the literature reviews, the relationship is not just a straightforward one. Teenagers tend to disclose more information than adults while apply privacy setting less strictly at the same time (Wlarave et al., 2012). Teenagers use three strategies to manage their own information: 1) limiting the level of disclosure, 2) using privacy settings, and 3) managing friends/audience, which work in conjunction with each other (Heirman et al., 2016). Certain platforms, such as TikTok, also use creative SMUC strategies, such as recognizing the nature of algorithms, sharing content on more private channels, or running multiple accounts (Ebert et al., 2023). However, the relationship between SMP and SMUC is complicated. Interestingly, teenagers who disclose private information more have more the friends, but less frequency of privacy setting use (Heirman et al., 2016). This indicates that a high level of self-expression does not necessarily mean that all kinds of reinforcement of management behavior. In addition, whether to disclose information or to keep it secret or hide depends on contextual factors such as individual beliefs about online risk or family rules (Babskie & Metzger, 2018). Cultural backgrounds or parents' intervention is also able to affect teenagers' behavior toward information disclosure (Laohabutr et al., 2021). As such, overall, it seems that the more online activities and self-expression can be seen to increase the need for management itself in some form, and this study seeks to verify the hypothesis that SMP will have a positive effect on SMUC.

The fifth hypothesis predicts that the higher the level of teenagers' digitally mediated social interaction and connection, the higher the negative self-conception related to social media. In other words, the higher the intensity of online interaction or social media use, the lower the self-esteem or the more likely they are to perceive themselves negatively, such as feeling fake, anxious, and alienated. This is mainly related to the society comparison or social alienation, which commonly occurs in an online environment (Liu, 2025; Hussain et al., 2025). Previous studies provide consistent evidence to support this negative relationship. The tendency of adolescents' self-esteem to decrease significantly as social media usage intensity or time increases has been repeatedly observed in several studies (Farooq et al., 2023; Hussain et al., 2025; Makhfudli, 2020), and

experience of social alienation in online interaction is important reason to decrease self-esteem (Hussain et al., 2025). In addition, recent social media usage decreases the current self-esteem in real time has been reported through experiential sampling studies (Miljeteig & Soest, 2022). The overall findings strongly suggest that certain aspects of PFSA (e.g., high intensity of use/time, interactions experiencing alienation) tend to increase SETB (low self-esteem, negative self-awareness). Therefore, this study attempts to establish and test the hypothesis that the PFSA will have a positive effect on SETB.

The sixth hypothesis predicts that the higher the level of social media-related negative self-awareness (SETB) in adolescents, the higher the severity perception (NC) of common problems or risks in the online environment. The previous studies indicate that the impact of negative self-awareness on risk perception (NC) is not simple but complex. Low self-esteem (SETB) can lead adolescents to engage in risky online behavior on a whim to gain peer recognition (e.g., participation in dangerous challenges), and serious consequences or risks can be overlooked in the process (Iskajyan, 2024; Aktu, 2024). On the other hand, if teenagers deliberately focus on their negative attributes or through the self-diagnosis process (Ling et al., 2012), self-positive bias can be reduced, and their judgment on "personal" risks (e.g., internet addiction risks) can be more realistic and accurate. Thus, direct empirical evidence between SETB and NC is lacking and the relationship is complex. Nevertheless, based on the initial theoretical possibility that negative self-awareness can increase 'sensitivity' to risk, this study seeks to explore and verify the hypothesis that H6: SETB will have a positive effect on NC. Drawing upon the hypotheses presented above, the baseline model illustrated in Figure 1 is proposed.

In the field of bullying research, Multiple Group Analysis (MGA) is actively used to identify differences in experience or psychological mechanisms between various groups. Previous studies applied MGA to compare and analyze how the pattern of bullying occurring in various contexts such as school, workplace, and cyberspace varies according to group characteristics such as gender, culture, and role within the organization (Malik et al., 2024; Park & Han, 2018). In addition, MGA was used as a key methodology to verify that the effectiveness of a particular intervention or the relationship structure between related variables remains the same between groups (measurement or structural identity) (Chen et al., 2020). As such, MGA is evaluated as an analysis tool that provides important insights for understanding in-depth the complex and multifaceted characteristics of bullying and developing group-specific intervention strategies.

Commented [RX2]: Six hypotheses are logically derived and richly supported by theory and empirical findings. The review is comprehensive and appropriately cautious in its framing.

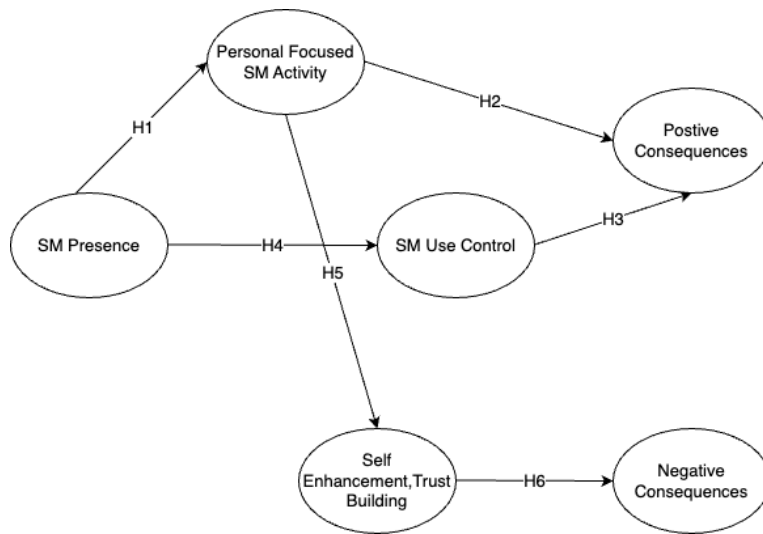


Figure 1. Baseline Model

Data Description

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This study collected data from the "Teens, Social Media & Technology 2018" survey of teenagers aged 13 to 17 in the United States from March 7 to April 10, 2018, implemented by the Pew Research Center. The survey was conducted by the NORC AmeriSpeak Panel, which is a probability-based panel that represents the U.S. household population. The survey was conducted through online and telephone interviews with teenagers.

Through the data purification process, a total of 743 teenagers' respondents were used in the final analysis of this study. The demographic characteristics of the final analysis sample (N=743) are as follows. By age, 301 people aged 13-14 (about 40.5%) and 442 people aged 15-17 (about 59.5%). The gender distribution was 348 boys (about 47.0%) and 393 girls (about 53.0%) (as of N=741 excluding 2 gender non-response). The main groups in the race/ethnic composition included 355 non-Hispanic whites, 129 non-Hispanic blacks, and 202. Regarding the the household income level, 199 (about 26.8%) were under \$30,000, 266 (about 35.8%) were with \$30,000 - \$74,999, and 278 (about 37.4%) were over \$75,000. The final educational background of the respondent's parents was also collected. 142 (about 19.3%) were under high school graduation, 265 (about 36.0%) had some college graduates, and 329 (about 44.7%) had college graduation or higher (as of N=736), excluding 7 non-response from parents).

In this study, cross-validation was conducted by dividing the entire sample (N=743) into a calibration sample (N=372) and a validation sample (N=371) to secure the reliability of the model. The main variables used in the analysis were measured on a Likert scale or a categorical response scale. There were no separate weights were applied in the SEM analysis of this study. Refer to Table 1 for detailed descriptions of the observation variables and the corresponding questionnaire questions.

The data used in this study are secondary data publicly provided by the Pew Research Center, and ethical issues in the data collection process are believed to have been managed according to the institution's procedures.

Table 1

Construct	Variable Name	Variable Description
Personal Focused SM Activity	FRIEND1	In a typical week, how often do you get together with friends...In person (outside of school or school-related activities)?
	FRIEND2	In a typical week, how often do you get together with friends...Online (including on your cellphone, on social media, or through online gaming)?
	FRIEND3	In general, how much time would you say you spend with your friends IN PERSON?
	FRIEND5	How many people would you consider to be your close friends?
	CELL2A	When you are using your cellphone, how often, if ever, are you...?
	CELL2B	When you are using your cellphone, how often, if ever, are you...?
	CELL3A	[Feel as if you have to respond to messages from other people immediately] Still thinking about your cellphone, how often, if ever, do you...?
	CELL3B	[Check for messages or notifications as soon as you wake up] Still thinking about your cellphone, how often, if ever, do you...?
	CELL3C	[Lose focus in class because you're checking your cellphone] Still thinking about your cellphone, how often, if ever, do you...?
	GROUP1	How often, if ever, do you spend time in any groups or forums online (including on blogs, social media, or online gaming forums)?
	POST2 (Countvar)	Which of the following things, if any, do you ever post about on social media?
SM Presence	FRIEND6	Are any of your close friends...?
	POST1A	[Selfies] How often, if ever, do you post the following things on social media?
	POST1B	[Videos you've recorded and created] How often, if ever, do you post the following things on social media?

	POST1C	[Updates on where you are or what you're doing] How often, if ever, do you post the following things on social media?
	POST1D	[Things that only your closest friends would understand] How often, if ever, do you post the following things on social media?
	POST1E	[Things you want to go viral] How often, if ever, do you post the following things on social media?
	SNS1 (Countvar)	Do you ever use any of the following social media sites?
Positive Consequences	SOC2POSA	[More in touch with your friends' feelings] In general, does what you see on social media make you feel...?
	SOC2POSB	[Like you have a place where you can show your creative side] In general, does what you see on social media make you feel...?
	SOC2POSC	[More connected to what's going on in your friends' lives] In general, does what you see on social media make you feel...?
	SOC2POSD	[Like you have people who can support you through tough times] In general, does what you see on social media make you feel...?
	GROUP3A	[Helping you get through tough times in your life] Thinking about online groups where you spend time, how much of a role, if any, have they had in...?
	GROUP3B	[Making you feel more accepted] Thinking about online groups where you spend time, how much of a role, if any, have they had in...?
	GROUP3C	[Helping you figure out how to feel about important issues] Thinking about online groups where you spend time, how much of a role, if any, have they had in...?
	GROUP3D	[Exposing you to new types of people] Thinking about online groups where you spend time, how much of a role, if any, have they had in...?
	SOC4 (Countvar)	Here are a few different pairs of words. For each pair, please select the one that most closely matches how you feel when using social media – even if neither is exactly right. Does social media make you feel more...
	SOC6	How often, if ever, do you unfriend or unfollow people on social media?
SM Use Control	SOC5A	[Organize your feed to only see certain types of content] Thinking about your experiences on social media... How often, if ever, do you do each of the following?
	SOC5B	[Delete or restrict access to things you post because you are concerned, they could negatively impact you later in life] Thinking about your experiences on social media... How often, if ever, do you do each of the following?
	SOC5C	[Delete or restrict access to things you post because you don't want your parents to see them] Thinking about your experiences on social media... How often, if ever, do you do each of the following?
	LIMIT (Countvar)	Have you ever cut back on the amount of time you spend...

Self Enhancement, Trust Building	SOCEXPB	[Find trustworthy information] How much, if at all, do you think social media helps people your age do each of the following?
	SOC4A	[Authentic/Fake] Here are a few different pairs of words. For each pair, please select the one that most closely matches how you feel when using social media – even if neither is exactly right.
	SOC4B	[Confident/Insecure] Here are a few different pairs of words. For each pair, please select the one that most closely matches how you feel when using social media – even if neither is exactly right.
	SOC4D	[Included/Excluded] Here are a few different pairs of words. For each pair, please select the one that most closely matches how you feel when using social media – even if neither is exactly right.
Negative Consequences	WORRYA	[On your cellphone] Overall, how much time would you say you spend...?
	WORRYB	[On social media] Overall, how much time would you say you spend...?
	WORRYC	[Playing video games] Overall, how much time would you say you spend...?
	OH1A	[People being harassed or bullied] Thinking of some experiences that might happen to people your own age when they use the internet or cellphones...
	OH1B	[Privacy violations such as identity theft or stolen information] Thinking of some experiences that might happen to people your own age when they use the internet or cellphones...
	OH1C	[People seeing false or inaccurate information] Thinking of some experiences that might happen to people your own age when they use the internet or cellphones...
	OH1D	[People spending too much time online] Thinking of some experiences that might happen to people your own age when they use the internet or cellphones...
	OH3 ^m	Still thinking about your experiences online or on your cellphone, which of the following, if any, has ever happened to you personally?

Note. m = Variable used only in Multiple Group Analysis. All factor loadings are standardized.

CFA/SEM Analysis

Confirmatory Factor Analysis (CFA)

Before the implementation of Structural Equation Modeling, confirmatory factor analysis (CFA) was conducted to evaluate the validity of the baseline model (figure 1) based on the six latent variables (PFSA, PC, SMP, SMUC, SETB, NC). For the analysis, Mplus software was used, and the maximum likelihood method was used.

The initial goodness-of-fit of this baseline model was evaluated for the entire sample (N=743). Some indicators reached acceptable levels, but the overall goodness-of-fit required

improvement (e.g., CFI = 0.640 for the initial model, RMSEA = 0.069, Model CFA 0). Therefore, based on the correction index and theoretical review, the measurement model was sequentially improved by removing the problematic loading items (e.g., high cross-loading, low coefficient loading). Table 2 details this iterative model modification process and the improvement of the goodness-of-fit index at each stage. As a result of evaluating the fit of the final model for the entire sample, $\chi^2 = 381.510$ ($p < .001$), and the other major goodness-of-fit indices were CFI = 0.949, TLI = 0.939, RMSEA = 0.040 (90% CI: 0.035-0.046). Overall, meeting the recommended criteria (e.g., CFI/TLI $> .90$, RMSEA $< .06 \sim .08$, SRMR $< .08$; Hu & Bentler, 1999), the measurement model in this study was judged to fit the data well.

Table 2 Model Modification Process for the model

Model	CFI	RMSEA	Variable Deleted	Raseon for Deletion Using Lagrange Multiplier Test
CFA 0	0.64	0.069	-	None: base model for CFA
CFA 1	0.836	0.057	SOCG, SOCEXPB, SOC4, LIMIT, WORRYA, WORRYB, WORRYC	Many measured variables on the lated factors
CFA 2	0.862	0.057	FRIEND1, CELL2B, GROUP1, FRIEND6	Many measured variables on the lated factors
CFA 3	0.943	0.04	GROUP3D, GROUP3A, GROUP3B	Deleted variables double-loaded on additional latent factors
CFA 4	0.946	0.041	GROUP3C, OH1D	Deleted variables double-loaded on additional latent factors
CFA 5	0.954	0.039	POST2	Deleted variables double-loaded on additional latent factors
CFA 6	0.959	0.038	SNS1	Deleted variables double-loaded on additional latent factors
CFA 7	0.958	0.037	FRIEND6(Revoke)	

In addition, to confirm the stability of the model, the same CFA model was applied to the calibration sample (N=372) and the validation sample (N=371), respectively, and analyzed. Model fit in the calibration sample $\chi^2 = 261.035$ ($p < .001$), CFI=0.958, TLI=0.950, RMSEA=0.037 overally satisfy the criteria, and the fitness in the validation sample was also $\chi^2 = 294.819$ ($p < .001$), CFI=0.939, TLI=0.926, RMSEA=0.043, which indicates that the indices maintained a good level. As the fitness index between the two sub-sample did not show a significant difference, stability of the measurement model was confirmed.

The final CFA model goodness-of-fit index for all samples is summarized in Table 3. To evaluate the quality of the final measurement model, standardization factor loading, composite reliability (CR), and mean variance extraction (AVE) were reviewed mainly based on the calibration sample results (Table 4). As shown in Table 4, the standardized factor loading (λ) of all measurement items was statistically significant ($p < .001$) and generally met the recommended level of 0.5 or higher (particularly within PFSA, some items exhibited low loading requiring careful interpretation). This supports the convergent validity of the items measuring each latent variable. Although the configuration reliability evaluated through CR and the convergent validity

evaluated through AVE mostly met the acceptable thresholds (CR >.70, AVE >.50), some configuration items showed slightly lower values than these criteria.

Table 3 Confirmatory Factor Analysis (CFA) Results: Model Fit and Measurement Quality

Sample	χ^2	df	p	CFI	TLI	RMSEA	90% CI	SRMR
Overall (N=743)	381.51	174	<.001	0.949	0.939	0.04	[0.035, 0.046]	0.038
Calibration (N=372)	261.035	174	<.001	0.958	0.95	0.037	[0.027, 0.046]	0.041
Validation (N=371)	294.819	174	<.001	0.939	0.926	0.043	[0.035, 0.052]	0.048

Table 4 Summary of Measurement Model Evaluation (CFA)

Construct	Indicator	Std. Loading (λ)	CR	AVE
PFSA	FRIEND2	0.555***	0.531	0.285
	CELL2A	0.646***		
	FRIEND6	0.358***		
PC	SOC2POSA	0.777***	0.821	0.535
	SOC2POSB	0.685***		
	SOC2POSC	0.697***		
	SOC2POSD	0.763***		
SMP	POST1A	0.647***	0.781	0.418
	POST1B	0.646***		
	POST1C	0.696***		
	POST1D	0.674***		
	POST1E	0.560***		
SMUC	SOC5A	0.536***	0.699	0.441
	SOC5B	0.676***		
	SOC5C	0.761***		
SETB	SOC4A	0.609***	0.671	0.406
	SOC4B	0.694***		
	SOC4D	0.605***		
NC	OH1A	0.805***	0.795	0.565
	OH1B	0.746***		
	OH1C	0.701***		

Overall, as a result of CFA for all samples and subsamples, the measurement model of this study showed that each item adequately measures the corresponding latent variable with good model fit. This means that a stable measurement basis has been established to verify the research hypothesis through structural model analysis (SEM).

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Structural Equation Modeling (SEM)

After confirming the validity of the model throughout CFI method, Structural Equation Modeling (SEM) was implemented to verify the research hypothesis. The validity of SEM model for the entire sample (N=743) was $\chi^2=416.876$ (df=182, $p < .001$), CFI=0.942, TLI=0.934, RMSEA=0.042(90% CI [0.036, 0.047]), which is acceptable as Confirmatory Factor Analysis results. To verify the stability of the model, the SEM was implemented on the calibration sample (N=372) and the validation sample (N=371). The result of the analysis based on the calibration sample ($\chi^2 = 289.351$, $p < .001$, CFI=0.948, TLI=0.941, RMSEA=0.040, SRMR=0.050) and the result of the analysis based on the validation sample ($\chi^2 = 313.574$, $p < .001$, CFI=0.934, TLI=0.923, RMSEA=0.044, SRMR=0.053), the stability of the model was confirmed.

As a result of verifying the path significance of the overall sample (N=743) using SEM model (see Table 5 for details), the factor loadings of the measurement model were all significant and remained good, as in the CFA results. Looking at the standardization coefficient (β) of the structural path, SMP had a significant positive (+) effect on PFSA ($\beta=0.615$; $p<.001$) and SMUC ($\beta=0.485$; $p<.001$), respectively. For PC, both PFSA ($\beta=0.806$, $p<.001$) and SMUC ($\beta=0.146$, $p=.002$) were found to have a significant positive (+) effect. In addition, PFSA had a significant negative (-) effect on SETB ($\beta=-0.327$, $p<.001$) and SETB had a significant positive (+) effect on NC ($\beta=0.194$, $p<.001$). The error covariance between the set PC and NC was not statistically significant ($p=.062$). The direction and significance of these path coefficients were generally consistent in the development and verification sample analysis, supporting the stability of the results.

As a result of the path analysis using the calibration and validation sample, the main paths that were significant in the entire sample (SMP→PFSA, SMP→SMUC, PFSA→PC, SMUC→PC, PFSA→SETB, SETB→NC) were found to be consistently statistically significant in both the development and verification samples. The size of the path coefficient also showed no significant difference between the three samples. However, the error covariance between PC and NC was significant in the development sample ($p=.032$), but not in the test sample ($p=.728$), showing a pattern similar to the overall sample result. These results suggest that the main paths of this study model appear stably despite the sample segmentation.

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Table 5: SEM Model Result

Hypothesis	Suggested Relationships	Estimated Loading	P-Value
H1: SMP → PFSA	+	0.704	0.000***
H2: PFSA → PC	+	0.784	0.000***
H3: SMUC → PC	+	0.144	0.035*
H4: SMP → SMUC	+	0.543	0.000***
H5: SETB → NC	+	-0.240	0.001**
H6: PFSA → SETB	+	0.232	0.001**

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Multiple Group Analysis

To further examine the structural stability of the proposed model across different experiences of online harassment and bullying, a multiple group analysis (MGA) was conducted. The grouping variable was based on the respondents' scores on the item "OH3," which measures the frequency of harassment and bullying experiences online. Respondents were divided into two groups, "more harassed/bullied" and "less harassed/bullied," using the median split method. The primary objective was to test whether the structural relationships in the model significantly differed between these two groups under the null hypothesis that no substantial differences exist.

The analysis followed a sequential procedure (see Table 6). First, measurement invariance across groups was assessed. Configural invariance, representing the baseline model with the same factorial structure across groups but freely estimated parameters, was first tested. Subsequently, metric invariance, constraining factor loadings to be equal across groups, was examined. The chi-square difference test comparing the metric model to the configural model yielded a p-value of 0.2682, indicating that the assumption of metric invariance could not be rejected. Thus, it was concluded that the factor structure and loadings were equivalent across groups, allowing for subsequent comparisons of structural path coefficients.

Following the establishment of metric invariance, the invariance of regression coefficients was tested. Initially, a fully unconstrained model, allowing all structural path coefficients to vary freely between groups, was estimated, and its chi-square value and degrees of freedom were recorded. Subsequently, a series of nested models were tested by constraining individual path coefficients to be equal across groups, one at a time. Each nested model was compared against the unconstrained model using the Satorra-Bentler scaled chi-square difference test. If the p-value was greater than 0.05, it was interpreted that the respective path did not significantly differ across groups; if the p-value was less than 0.05, it indicated a significant group difference.

The Wald test results revealed that among all structural paths, only the path from PFSA to SETB showed a significant group difference. Specifically, the p-value associated with the equality constraint on this path was 0.0432 (less than 0.05), indicating that the effect of PFSA on SETB significantly differed between the "more harassed/bullied" and "less harassed/bullied" groups. Further examination of the estimated standardized path coefficients revealed that, while both groups exhibited a negative relationship between PFSA and SETB, the magnitude of this negative effect was stronger in the "less harassed/bullied" group. This suggests that among adolescents with fewer harassment or bullying experiences, increased social media interaction is more strongly associated with reductions in negative self-perceptions. In contrast, for adolescents who experience higher levels of harassment or bullying, the protective effect of social media interaction against negative self-conception appears to be weaker.

These findings highlight the nuanced role that online harassment and bullying experiences may play in moderating the relationship between social media engagement and psychological

outcomes. While positive interactions through social media generally mitigate negative self-conceptions, the buffering effect is less pronounced for adolescents who are more frequently exposed to online harassment.

Commented [RX6]: Slightly more theoretical reflection on why this moderation occurs (e.g., emotional exhaustion, dissonance, coping resources) would deepen the implication.

Table 6: MGA Analysis Results

Testing for measurement invariance

Models Compared	Chi-Square	Degrees of Freedom	P-Value
Metric against Configural	17.895	15	0.2682
Scalar against Configural	51.213	30	0.0092**
Scalar against Metric	33.318	15	0.0042**

Testing for structural invariance

Hypothesis	Path	Wald Test P-Value	Results
H1a-H1b=0	SMP → PFSA	0.7432	Not supported
H2a-H2b=0	PFSA → PC	0.0781	Not supported
H3a-H3b=0	SMUC → PC	0.9451	Not supported
H4a-H4b=0	SMP → SMUC	0.6257	Not supported
H5a-H5b=0	PFSA → SETB	0.0432*	Supported
H6a-H6b=0	SETB → NC	0.2494	Not supported

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Validation

As previously discussed in the sections on CFA and SEM, to evaluate the stability and generalizability of the measurement and structural models, cross-validation analyses were conducted using a calibration sample and an independent validation sample. Both the CFA model and the SEM model were examined across three conditions: calibration, validation without fixed estimates, and validation with fixed estimates. Model fit was assessed using chi-square (χ^2), Comparative Fit Index (CFI), Root Mean Square Error of Approximation (RMSEA) and its 90% confidence interval (CI), and Schwarz's Bayesian Information Criterion (BIC) (see Table 7 for details).

In the calibration sample (df =174), the base model demonstrated an excellent fit to the data. When validated without fixed parameter estimates, the model maintained an acceptable fit. Although a slight decline in model fit was observed when parameter estimates were fixed, the fit indices remained within acceptable thresholds, supporting the overall stability and replicability of the model structure across samples. Additionally, the revised model, incorporating theoretically motivated modifications to improve parsimony, was subsequently evaluated. In the calibration sample (df =182), the revised model achieved good fit indices. Validation without fixing parameter estimates indicated a stable model fit. Furthermore, when estimates were fixed in the second validation sample, the revised model exhibited better generalizability compared to the base model.

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Overall, the results of the cross-validation analyses provide evidence for the robustness and generalizability of both the measurement and structural models. The baseline model, assessed using CFA, yielded acceptable and stable fit indices across calibration and validation samples, supporting the adequacy of the measurement structure. Based on this foundation, the revised model, estimated using SEM, further demonstrated consistent model fit and stronger generalizability, particularly under the more stringent condition of fixed parameter estimates. These findings suggest that the hypothesized structural relationships are well supported and that the model structure is replicable across independent samples.

Table 7: Cross-Validation Results

Model	Statistic	Calibration	(Without Fixed Estimated)	(With Fixed Estimates)
			Validation 1	Validation 2
CFA Model	df	174	174	252
	Chi-square	2293.331	2189.119	2189.119
	CFI	0.958	0.939	0.902
	RMSEA	0.037	0.043	0.046
	RMSEA 90% CI	(0.027, 0.046)	(0.035, 0.052)	(0.039, 0.052)
	Schwarz criterion	16771.057	16693.186	1638.012
SEM Model	df	182	182	246
	Chi-square	2293.331	2189.119	2189.119
	CFI	0.948	0.934	0.911
	RMSEA	0.04	0.044	0.044
	RMSEA 90% CI	(0.031, 0.048)	0.036	0.037
	Schwarz criterion	16752.021	16664.611	16394.238

Note: In the base model, there are more degrees-of-freedom in the calibration sample than in the validation sample without fixed estimates, because a constraint was set for convergence in the calibration sample. This constraint is sample specific and was released with estimating the model in the validation sample.

Discussion

This study investigated the structural relationships between teenagers' patterns of social media use and their psychological outcomes, using nationally representative survey data. Overall, the findings offer strong empirical support for the proposed model. Specifically, five out of the six hypothesized relationships were confirmed. Teenagers' social media posting behavior (SMP) significantly predicted both digitally mediated social interaction (PFSA) and content management efforts (SMUC). In turn, both PFSA and SMUC were positively associated with adolescents' perception of the positive outcomes (PC) of social media use. Additionally, higher levels of negative self-perception (SETB) significantly predicted greater recognition of online risks (NC). These findings align with previous research indicating that active social media engagement facilitates social connection and that heightened negative self-awareness sensitizes individuals to potential digital risks (Pew Research Center, 2018a; Pew Research Center, 2018b; O'Reilly et al., 2022).

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A key exception emerged regarding Hypothesis 5. Despite the complex and mixed findings in prior literature concerning the relationship between the two variables, with interpretations ranging from positive to negative associations, the data indicated a significant negative relationship: teenagers who engaged more in social media interactions reported lower levels of negative self-perception. Several contextual factors may explain this finding. First, the timing of data collection must be considered. The survey was conducted in early 2018, during a period when adolescents' use of platforms such as Instagram, Snapchat, and short video features was characterized by active participation, creative self-expression, and peer feedback. Rather than passive consumption, adolescents were engaging in ways that encouraged positive identity development and community building (Pew Research Center, 2018d). Previous research has highlighted that active, meaningful social media interactions can foster self-esteem and subjective well-being, rather than undermine them (Ni et al., 2019; Pew Research Center, 2018a).

Second, sample characteristics and increasing digital literacy may have played a role. By 2018, awareness of digital risks had grown significantly, alongside greater emphasis on parental involvement and structured digital literacy programs (Pew Research Center, 2018c). Adolescents exposed to these supports may have been better equipped to manage online risks and navigate social media in healthier ways, reducing the likelihood that high engagement would harm their self-perception.

Third, biases related to self-reporting should be acknowledged. Adolescents experiencing more positive social media outcomes may have been more willing to participate and disclose favorable experiences, while those facing greater difficulties may have been underrepresented or may have minimized their responses, consistent with known patterns in survey research.

Moreover, a growing body of literature supports the idea that positive and authentic self-presentation online can enhance adolescents' self-esteem and self-awareness. Engaging in creative self-expression, receiving positive peer feedback, and actively managing one's digital persona have been linked to stronger self-esteem and healthier identity development (Hossain & Wahid bin Ahsan, 2024; Liu, 2025; Yang, Kwon, & Jang, 2023). Such experiences can foster reflection on personal traits and social roles, contributing to psychological growth rather than deterioration (Wang, 2024; Ni et al., 2019).

These findings suggest that the relationship between social media use and adolescent well-being is multifaceted and context dependent. Rather than being universally harmful, social media interactions may serve as important resources for psychological resilience and positive identity formation under the right conditions. Such as supportive peer environments, higher digital literacy, and proactive self-management.

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Contribution and Limitations

This study offers a comprehensive examination of how teenagers' social media usage relates to both positive and negative psychological outcomes. Drawing on a large-scale, nationally representative sample, it tested a structural model that integrates key aspects of online behavior, including posting activities, social interaction, content management, and outcome perceptions. The model was largely supported by the data, with five of six hypothesized relationships confirmed. By multi-group analysis, the study further identified how experiences of online harassment may moderate these processes, adding depth to existing discussions around social media and adolescent development.

One of the most notable contributions of this study is its finding regarding the relationship between digitally mediated social interaction (PFSA) and negative self-perception (SETB). Contrary to expectations, greater engagement was associated with lower levels of negative self-awareness. This result challenges a dominant narrative in the literature that emphasizes the harmful effects of online interaction, suggesting instead that, under certain circumstances, social media participation may foster more resilient self-concepts among adolescents. Given the timing of data collection and the evolving landscape of social media use, this finding points to the importance of context when interpreting digital behaviors.

However, several limitations should be considered when evaluating these results. First, the cross-sectional nature of the data limits the ability to draw causal inferences. The observed associations, while theoretically grounded, cannot establish directionality. Second, the reliance on self-reported measures may introduce biases, including social desirability effects and recall inaccuracies. Third, although the sample is nationally representative of U.S. adolescents, the findings may not generalize to teenagers in other cultural settings where social media usage norms differ. Lastly, the method of dividing participants into groups based on the median value of harassment experiences, while practical for analysis, may have oversimplified the variability within those experiences.

Research Directions

Future studies should consider longitudinal approaches to better capture how teenagers' online behaviors and psychological outcomes evolve over time. Such designs would allow for a clearer understanding of causality and the temporal dynamics between social media engagement and self-perception. It will also be important to move beyond measuring the volume of social media use toward examining the quality of engagement. Distinguishing between active, meaningful interactions and passive, potentially harmful behaviors may yield more precise insights into which aspects of digital life are beneficial or detrimental to adolescent well-being. Expanding the research across different cultural contexts is another important direction. Social media platforms are used differently across societies, and the protective or harmful effects observed in this study may not be universal. Comparative studies could help clarify whether the patterns found here are specific to U.S. adolescents or more widely applicable. Additionally, examining the role

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of protective factors, such as digital literacy skills, family support, and individual coping strategies could deepen understanding of why some teenagers benefit from social media engagement while others are more vulnerable to its risks. Finally, qualitative research that captures teenagers' perspectives in their own words could offer valuable insights into how they navigate social media, manage their online identities, and make sense of both the opportunities and challenges that digital spaces present.

Commented [SB11]: I read your report carefully. I am convinced that you benefited immensely from the group-specific online sessions during various phases of this research project. It is clear that you have learned key aspects of effective execution of the SEM method. Congratulations.
Ruiqing has offered very thoughtful comments above, so I will not reiterate those here. I think your team has progressed a good deal toward my objective to imbue the ability in you to conduct an SEM investigation successfully from scratch for your dissertation (if needed). Congratulations.
Overall, this is a very good report. Your group's grade for this research project is as follows:
330/350

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